

COMP 370 Data Science Project - Politician in the Media

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Introduction

This study examines recurring themes and sentiment patterns in North American media coverage of Donald Trump during October and November 2025, which is a critical period following the 2025 elections. We address the following research question: *What recurring themes emerge in media coverage of Donald Trump, and what sentiment patterns can be observed across these topics?*

Collected articles for this time period cover very recent news around specific controversies: the Epstein files saga, the Trump-Mamdani meeting and Saudi-Khashoggi relations. Our findings suggest that the media frames Trump's presidency as controversial and not always very positive. Through open coding, sentiment annotation, TF-IDF analysis, and LLM-generated summaries, we systematically analyze media coverage of Donald Trump to answer this question.

Data

Data Collection

We started by collecting a preliminary sample of 1600 news articles published between October and November 2025 using the World News API. An article is selected as a candidate for this first dataset if the keyword "Donald Trump" appeared at least once in either its body or title and was published by an English North American news outlet. The candidate is added as a member if it had a title unique to all articles already included.

When collecting these initial queries from the API, we kept the following fields for each article: the article's title, the source URL, the extracted opening (described below) and the sentiment score provided by the API (ranging from -1 to 1). For every article that passed the first round of filtering, we created a dataset row containing these fields in that order.

During the first round of filtering, we extracted the query article's first four sentences (what we would consider to be the "opening" of an article). After obtaining the initial dataset, we verified each entry for the appearance of the "Donald Trump" keyword in its opening. An article is added

to our second dataset if it passes all previously mentioned checks.

After this, we performed a manual overview of the second dataset. We started by removing all articles that are not beneficial to our study (exact duplicates, non-English articles). Next, we manually read through each entry's opening and removed it if it did not give us the impression that the main focus of the article was Donald Trump. After these rounds of filtering, the final dataset contained 550 entries.

Data Sampling and Bias

We collected our articles using the World News API. This API returns queries based on what is available at the time of data collection. In other words, we employed convenience sampling. Since we employed convenience sampling, the data collection method was not random, which means that inherent bias may have been introduced into the dataset. The API's query method might of had introduced bias into our dataset, since articles published around the time that data collection was done had a higher probability of being included.

Although a recency bias is usually undesired, the previously mentioned bias aligns with our target population and is consistent with our objective. We were tasked with determining the sentiment surrounding Donald Trump in articles published between October and November 2025. Because of this, the API's bias towards recent articles does not hinder our process, instead it aligns perfectly with the time window we are interested in studying. While the resultant conclusions are not appropriate for describing Donald Trump's overall long-term portrayal in media, it is representative of the timeframe we care about examining; therefore, the obtained conclusions are adequate.

Methods

Topic Design

Through open coding of 200 articles, we read titles and openings to identify dominant topics in Trump coverage. By documenting emerging themes, we were able to consolidate overlapping categories and refine into six distinct topics: Administration, Legal Issues, Immigration Policy, Electoral Politics, Foreign Relations and Economic.

Legal Issues focused specifically on legal controversies and judicial challenges. This includes the Jeffrey Epstein files saga. Economic centered on tariffs, inflation and affordability. This also includes cost-of-living discussions, and voter reactions to economic conditions. Administration encompasses the day-to-day operations of Trump’s presidency, including governance style, personnel management, and interpersonal dynamics with other public figures both in-person and online. This category captures controversial rhetoric toward lawmakers and journalists, staff appointments and resignations, departmental reorganizations, and interactions with political allies and opponents. Foreign relations were about international diplomacy, foreign policy decisions, and relationships with other nations. Electoral politics covers election conflicts, campaigns, party dynamics, and electoral strategy. Immigration policy captures the coverage of Trump’s enforcement actions and policy changes affecting legal and illegal immigration. Articles in this category discuss deportation operations, temporary protected status decisions, ICE activities, and border security measures.

To minimize subjectivity, we established decision rules for boundary cases. Many articles covered the same events, but from different angles. For example, conflicts between Trump and other public figures were coded as administration unless the conflict involved formal court proceedings. In that case, they were coded as legal issues instead. Also, immigration foreign policy (e.g., refugee diplomacy) was coded as foreign relations if international negotiations were the primary focus. If the focus was instead on domestic enforcement within the United States, then the annotation remained immigration policy.

Full Data Topic Annotation

After establishing these topic definitions, we coded the entire dataset. Articles that genuinely spanned multiple topics were assigned to the single most dominant theme based on headline and opening paragraph by applying the decision rules established during topic design.

Following full data annotation, we separated articles by topic and generated representative summaries using Claude (Anthropic’s LLM) to synthesize patterns across each category. Through iterative refinement of prompts, we obtained coherent summaries characterizing each topic’s dominant narratives.

Topic Characterization TF-IDF

To compute the TF-IDF scores, we started by tokenizing the words in the articles’ opening into individual row entries and entries that corresponded to common stop-words (for instance, “the”, “a” and “an”) were removed. We then used the TF-IDF algorithm found in the *tidytext* package in R to calculate the scores for all remaining words. After computing the values, we grouped them by annotated topic and selected the top ten highest scoring words for each topic.

Sentiment Coding

To determine the sentiment around the political figure Donald Trump, we had to do a second round of manual coding,

where we labeled each article by positive, negative or neutral. This annotation was based on the title as well as the opening of each article. We have established a clear distinction between the three different labels. Articles were coded negative when the title or the opening talk about a harmful or unfavorable developments like wars, conflicts, policies that could harm people, military deployment, or even any intonation that clearly denoted a negative tone. For positive articles, the title and opening describe a favorable action or comment like tariff reductions, peace agreements, successful negotiations, a successful meeting or any explicitly positive comment on a person or a situation. For article that are considered neutral, they are the ones where the content is factual updates, quotations of political figures or a description of a situation without an opinion. In ambiguous situation, we would annotate this article as neutral because there is not enough proofs to say that it is neither positive nor negative and doing so we minimize over interpretation of the articles.

Results

Topics and LLM summary

Legal Issues. The Jeffrey Epstein files controversy dominated this topic (50% of coverage), chronicling Trump’s reversal from transparency promises to opposition before ultimately signing legislation after a 427-1 House vote. Coverage revealed a consistent pattern of institutional checks prevailing over executive authority, with courts blocking orders on birthright citizenship and DC National Guard deployment while Congress forced transparency. The bipartisan Epstein bill passage represented rare political unity driven by survivor advocacy.

Immigration Policy. Coverage focused on aggressive enforcement including mass deportations in Democratic cities, Somali TPS termination, and resumed ICE courthouse arrests, alongside military deployment for immigration enforcement. A recurring tension emerged between Trump’s hardline agenda and institutional pushback from federal judges who blocked initiatives and found Texas redistricting racially discriminatory. Despite increased fear, immigrant communities maintained resolve to remain.

Foreign Relations. Media concentrated on three storylines: Trump embracing Saudi Crown Prince MBS despite Khashoggi murder concerns while announcing massive arms deals, aggressive Venezuela posturing with military buildup, and a controversial Ukraine peace plan requiring territorial concessions to Russia and NATO abandonment. The coverage pattern revealed transactional diplomacy prioritizing bilateral deals over human rights and multilateral cooperation, evidenced by Trump’s G20 boycott and European ally marginalization.

Electoral Politics. Following Democrats’ strong 2025 off-year victories, coverage examined Trump’s aggressive redistricting efforts in Texas facing federal court blocks for racial gerrymandering, Republican concerns about Hispanic voter erosion, and Trump’s inflammatory rhetoric calling lawmakers “traitors” deserving death. The pattern showed Trump using executive pressure to reshape electoral maps

while Democrats leveraged his controversies for mobilization, with both parties facing coalition challenges ahead of crucial 2026 midterms.

Economic Policy. Coverage centered on Trump’s proposed \$2,000 “tariff dividend” checks facing Congressional resistance, selective Brazilian tariff rollbacks aimed at grocery relief, and growing voter disillusionment with supporters saying affordability has “only gotten worse.” Economists warned of stagflation signs despite Trump claiming an economic “sweet spot.” The pattern revealed reactive messaging alternating between success claims and policy reversals under political pressure, while voters increasingly expressed skepticism about delivery on affordability promises.

Administration. Coverage was dominated by Trump threatening Democratic lawmakers with death for “seditious behavior,” prompting 24/7 Capitol Police security, and Rep. Marjorie Taylor Greene’s resignation after being branded “Marjorie Traitor Greene” over Epstein disagreements. Trump’s surprisingly cordial meeting with NYC Mayor-elect Mamdani contrasted his previous “Communist Lunatic” characterization, while his “quiet, piggy” journalist insult was defended as “frankness.” Recurring patterns included extreme rhetoric requiring official reinterpretation, transactional political relationships, and steady erosion of presidential norms.

Topic Distribution

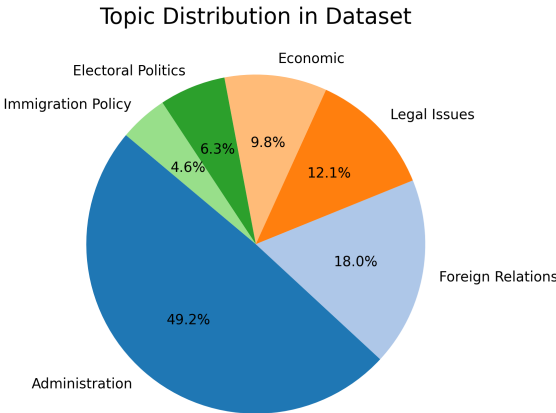


Figure 1: Pie chart representing the percentage of the topics in the dataset.

We can see in the pie chart that the ‘Administration’ topic represent nearly half of the dataset (49.2%) followed by the ‘Foreign Relations’ topic that have a percentage of 18.0%. For the less represented topics we can talk about ‘Immigration Policy’ and ‘Electoral Politics’ that are respectively represented by 4.6% and 6.3%.

Top 5 words for each topic TF-IDF

These 6 horizontal bar plots represent the top 5 results of the TF-IDF which show the top 5 words that are the

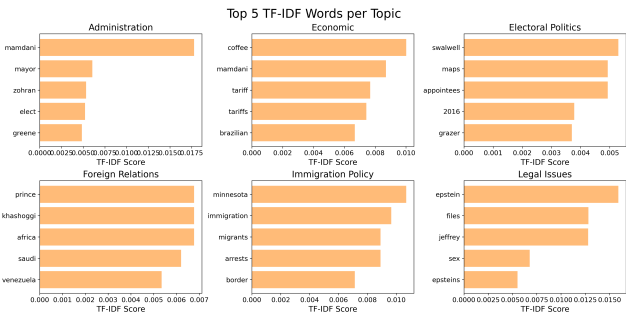


Figure 2: Top 5 TF-IDF words per topic.

most distinctive for each topic, we can see an emphasis on some words for each topic that describe the content of their belonging group. We can see the word ‘Mamdani’ with score of 0.0175 for the ‘Administration’ or even the words ‘prince’, ‘khashoggi’ or ‘africa’ that have the same score approximately 0.0065 for the ‘Foreign Relations’ topic. In this chart, we do not plot the 10 words for each topic for readability purposes.

Sentiment distribution

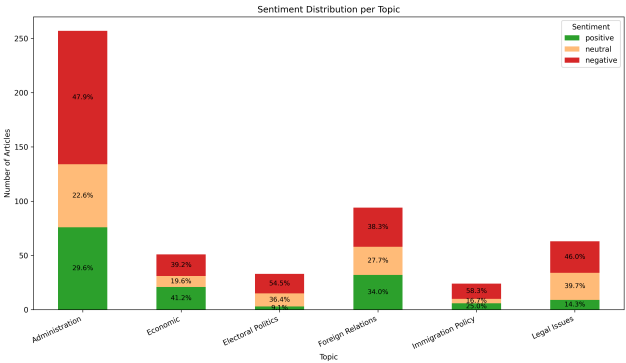


Figure 3: Bar chart representing the percentage of the sentiment per topic.

In this bar plot we can clearly see that the negative articles are the most present between the sentiment. In almost every topic the negative articles form the majority, the most flagrant are in topics ‘Immigration Policy’ with 58.3%, ‘Electoral Politics’ with 54.5%, ‘Administration’ with 47.9% of negative sentiments on the articles. The positive sentiment is the second best represented in the data set but in some topic it is over represented with 42.2% in the ‘Economic’ topic but also in the ‘Foreign Relations’ where the positive sentiment is not the most represented but represent a large proportion with 34.0%.

Discussion

Our analysis reveals a persistent theme in North American media coverage of Trump between October and November 2025: across all coded topics, his presidency is defined by

controversy rather than presidential achievements. In fact, the 'Administration' topic dominance (49.2% of coverage) from Figure 1 demonstrates that Trump's governing style itself became the story. From Figure 2, the high TF-IDF score for "mamdani" indicates that the media focused a lot of emphasis on this White House meeting because of interpersonal drama rather than policy substance. Referencing the LLM summaries across topics, coverage consistently mentions Trump's rhetoric, personnel conflicts, and norm-breaking behavior.

From Figure 1, the next two biggest shares go to 'Foreign Relations' and 'Legal Issues', which both emphasize controversial narratives over substantial outcomes. For example, foreign relations TF-IDF terms like "saudi," "prince," and "khashoggi" in Figure 2 all reference Trump's meeting with the Saudi crown prince for diplomatic developments, yet these were consistently contextualized against human rights. Similarly, articles related to 'Legal Issues' evidenced by TF-IDF scores where "epstein," "file," and "jeffrey" far outpace other terms in Figure 2, concentrated on the Epstein files saga which clearly shows that Trump's presidency attempted executive action to release those files as promised, but met institutional resistance.

Across these dominant topics and LLM summaries, media framed Trump's presidency through interpersonal confrontations, institutional battles, and controversial relationships. The coded topics themselves represent domains of presidential activity, but our results within each reveal coverage around controversy rather than legislative accomplishments or policy implementation.

Building on that topic structure, we can now establish a sentiment analysis that revealed a difference in how the media outlets evaluated Donald Trump through these different topics during October and November 2025. In almost every topic the negative articles form the majority, this show that the overall tone of the media during this period had a negative trend, but this negativity is not represented the same throughout all the subjects.

The topic which has the most negative articles is the 'Administration'. From Figure 3, we can see that the negative article represent 47.9% in the 'Administration' subject which represent at least 120 articles, against 29.6% for positive and 22.6% for neutral, which can be explained by the controversial article wrote in this topic where it's several articles talking about death threats, the clash with the president of the united states over the unreleased Epstein files or even some direct attacks on journalist. These contents portray the 'Administration' topic and explain why nearly half of the article have a negative sentiment. But we could also talk about the positive sentiment found in this section, where a lot of articles are related to the meeting between Trump and Mamdani, we can see that in the Figure 2, which show the word 'mamdani' having the biggest TF-IDF score for the 'Administration' topic. Before this event, there were a lot of articles talking neutrally even negatively about it. Against the odds, it happened to be a good reunion, so the tone changed positively, and a lot of articles were posted. The 'Legal Issues' show a similar trend where the negative sentiment is dom-

inant representing nearly half of the category with 46.0%, from Figure 3. Looking at most of the content spread by these types of articles we can see why the dominant sentiment is negative. The most recurrent topic was the Jeffrey Epstein files controversy where the TF-IDF words in this topic, from Figure 2, are only related to this news.

We can also see this negative trend in the 'Electoral Politics' and 'Immigration Policy' topics that are very small categories, but the negative articles represent more than half of the total number of articles in these subjects respectively 54.5% and 58.3%, from Figure 3. Following the trends of the other topics, the coverage in these two categories is very controversial: The republican concerns about Hispanic voter erosion and Trump calling lawmakers traitors deserving death for the 'Electoral Politics' field, or even the ICE courthouse arrests and military deployment for immigration enforcement for 'Immigration Policy' field. These matters support the controversial media coverage and explains the superior number of negative sentiments in these two topics.

In contrast, the 'Foreign Relations' topic show a more balance sentiment profile. From Figure 3, we still see that the negative sentiment is leading with 38.3% but we also have the positive sentiment that represent a good portion of this topic with 34.0%. This reflect the mix nature of the coverage in this topic, where some articles were about the Ukraine peace plan with Russia that often created a positive sentiment, but other coverage like the military build-up in Venezuela or Trump's G20 boycott created this negative sentiment trend that we have in the other fields. The other topic that stands out from the rest is 'Economic', where for once we have a majority of positive sentiments with a ratio of 41.2%, from Figure 3. The articles in this section talked the most about tariffs rollback on coffee in Brazil as we can see in Figure 2 with 4 out of the 5 words (coffee, tariff, tariffs, Brazilian) received the highest TF-IDF score which resonated with a positive sentiment. But we cannot neglect the presence of the negative sentiment with 39.2% of the category. These negative articles are talking about voters who thinks that affordability has gotten worse or economist that are alarmed by the president's decisions.

The goal of this study was to determine the prominent topics and sentiment patterns surrounding Donald Trump in North American news outline between October and November 2025. We concluded that the dominant sentiment during this time window was negative, as five of the six annotated topics, including the "Administration" topic which accounts for 49.2% of our dataset, had a majority of negative articles. However, 41.2% of articles in the "Economic" topic had a positive sentiment, while 39.2% were negative, making it the only topic in which positive sentiment formed the majority. News outlets prefer to highlight Donald Trump's controversies, instead of his administrative affairs. In fact, some articles even embed discussions about controversial topics such as human rights in articles focusing on diplomatic matters. During this time window, Donald Trump was focused on international activities such as the tariff war and the Ukraine peace plan which led to a substantial number of articles covering the "Foreign Relations" and "Economic" topics.

Nonetheless, the “Administration” and “Legal Issues” topics represent the big majority of the dataset, as internal affairs such as Jeffrey Epstein’s files and Zohran Mamdani’s election were the headline-driving and hot topics dominating the news during the data collection phase.

Group Member contributions

All group members equally participated in the successful completion of this project. We had weekly meetings to track the progress of the project while putting deadlines that align with our respective schedule. For individual work, the tasks were splitted equally and this is what each one of us did:

- **Mark:** Worked on the second round of annotation, the sentiment annotation. Also extracted the result from the data by producing plots and summarizing these results in text. Understood the situation to develop insights on the sentiment distribution to answer the problem question.
- **Sarah:** Worked on topic design. Also worked on the first round of annotation, the full data topic annotation, and LLM-generated summaries. Understood the situation to develop insights on the topic distribution and TF-IDF results to answer the problem question.
- **Zheng Ye:** Worked on data collection phase which includes script writing for the data collection and the TF-IDF calculations. Also, found the appropriate API for data collection and prepared the dataset used in the following steps.